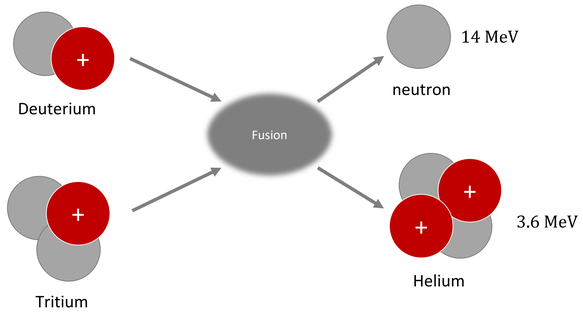
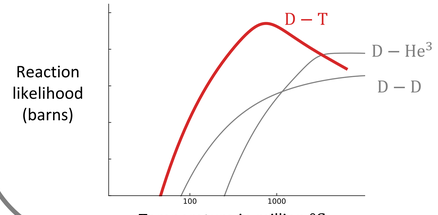
# ARC

## Affordable Robust Compact tokamak

**Concept from:**
Sorbom et al (2015), Fus. Eng. Design,
DOI: 10.1016/j.fusengdes.2015.07.008

## What is Deuterium-Tritium fusion?



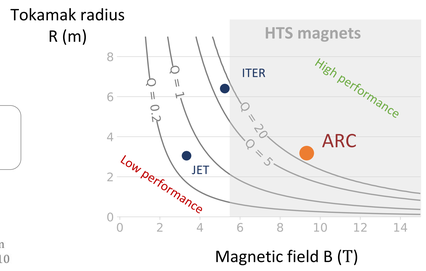



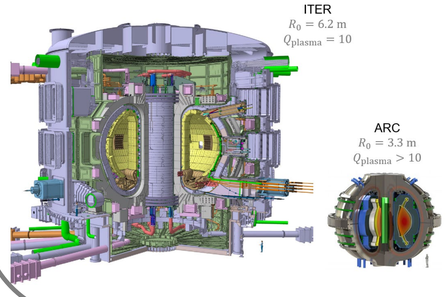

← Why do we need Deuterium and Tritium?

## High Temperature Superconducting Magnet

Increasing the magnetic field greatly improves fusion performance

$$Q_{\text{plasma}} = \frac{\text{fusion power}}{\text{heating power}} \propto R^3 \boldsymbol{B}^5$$

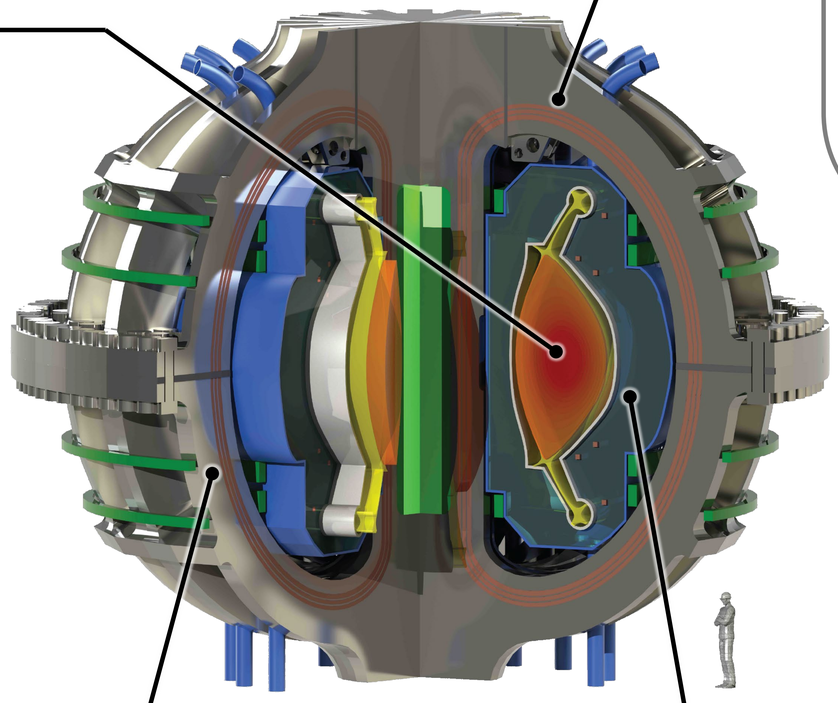






Using HTS magnets allows tokamaks to be more compact and cheaper



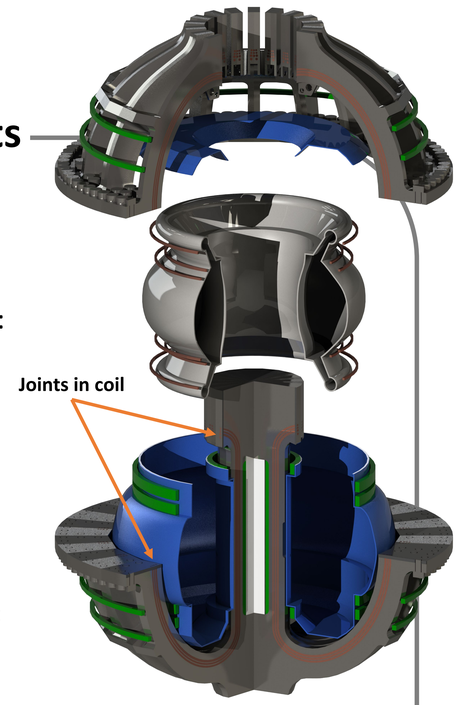


| | |
|---|---|
| Major radius | 3.3 m |
| TBR | 1.1 |
| Fusion power | 525 MW |
| Multiplication factor $Q$ | $> 10$ |
| Magnetic field | 9.2 T |

## Demountable magnets

Using **H**igh **T**emperature **S**uperconductors enables joints:

- ✅ Magnets can be mounted/demounted many times
- ✅ Easier maintenance from the top
- ✅ Vacuum vessel can be replaced and is no longer a lifetime component




**ARC's assembly video**

## Liquid Immersion Blanket

✅ Breeding tritium ✅ Shield from neutrons ✅ Extract heat

**Liquid Immersion Blanket**

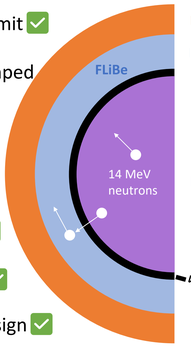
- Atmospheric pressure ✅
- No damage limit ✅
- Can be pumped entirely ✅
- Very low solid fraction ✅
- Effective moderator, breeder, heat sink ✅
- No gaps ✅
- Simple design ✅

High TBR ✅

**Conventional segmented steel blanket**

- ❌ Operates in vacuum
- ❌ Structural materials damaged by neutrons
- ❌ Has to be assembled inside the vacuum vessel
- ❌ High solid fraction
- ❌ Gaps leak neutrons
- ❌ Complex engineering

❌ Low TBR (for same blanket volume)




Why Lithium?

$$\mathrm{D} + \mathrm{T} \rightarrow \mathrm{He} + \mathrm{n}$$
$$\mathrm{Li} + \mathrm{n} \rightarrow \mathrm{T} + \mathrm{He}$$

Why Beryllium?

$$^{9}\mathrm{Be} + \mathrm{n} \rightarrow {}^{8}\mathrm{Be} + \mathrm{n} + \mathrm{n}$$

$$\textbf{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Theoretical sufficiency → TBR = 1
Losses + start other plants → $\mathbf{TBR \approx 1.1 - 1.2}$